Settimana: 21

Materia: Matematica
Classe: 5A
Data: 02/03/26

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$$\int x^2 \sin x \, dx \overset{\text{Per Parti}}{=}$$

$$\int f'g = fg - \int fg'$$

Goal: abbassare il grado di x^2

$$-x^2\cos x + \int 2x \cos x \, dx =$$

$g = x^2$ $\quad g' = 2x$

$f = -\cos x$ $\quad f' = \sin x$

$$-x^2\cos x + 2x\sin x - \int 2\sin x \, dx =$$

$$\boxed{-x^2\cos x + 2x\sin x + 2\cos x + c}$$

$g = 2x$ $\quad g' = 2$

$f = \sin x$ $\quad f' = \cos x$

$2\sin x\cos x = \sin 2x$

438: $$\int 8x \sin x \cos x \, dx = \int 4x \sin 2x \, dx$$

$2x = t$
$2dx = dt$

$$= \int t \sin t \, dt = -t\cos t + \int \cos t \, dt$$

$g = t$ $\quad g' = 1$

$f = -\cos t$ $\quad f' = \sin t$

$$= -t\cos t + \sin t + c$$

$$= \boxed{-2x\cos 2x + \sin 2x + c}$$

$$\int f'g = fg - \int fg'$$

Hard: $$\int \cos^2 x \, dx = \int \cos x \cdot \cos x \, dx =$$

$f = \sin x$ $\quad f' = \cos x$
$g = \cos x$ $\quad g' = -\sin x$

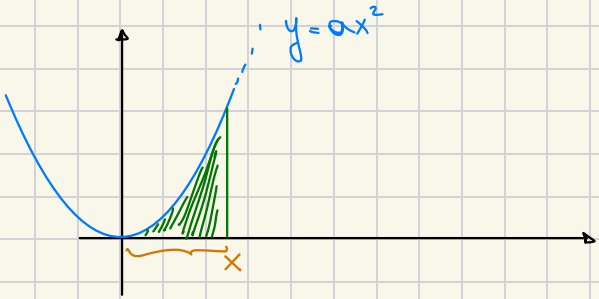
$$\sin x \cos x + \int \sin^2 x \, dx$$

$$\int \cos^2 x \, dx = \sin x \cos x + \int \underbrace{\sin^2 x}_{1-\cos^2 x} \, dx$$

$$\underbrace{\int \cos^2 x \, dx}_{R} = \sin x \cos x + \int 1 \, dx - \underbrace{\int \cos^2 x \, dx}_{R}$$

$$R = \sin x \cos x + x - R$$

$$\cancel{2}R = \boxed{\frac{\sin x \cos x + x}{2} + c = \int \cos^2 x \, dx}$$



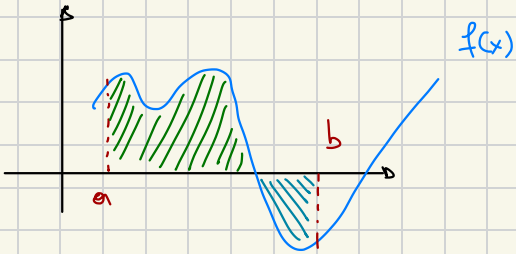
Area Verde $= \frac{ax^3}{3}$

$$\int ax^2 \, dx = \frac{ax^3}{3}$$

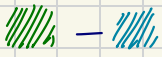
Teorema: Data $f : [a;b] \longrightarrow \mathbb{R}$ funzione integrabile, l'integrale

$$\int_a^b f(x) \, dx = F(b) - F(a)$$

con F primitiva di f è l'area sottesa dal grafico e dell'asse x



$$\int_a^b f(x) \, dx =$$



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$$f(x) = \frac{e^x(x-1)}{x^2}$$

1) Dom f : $x \neq 0$

2) Assi : $x = 0$ Impossibile

$y = 0$ $\quad \frac{e^x(x-1)}{x^2} = 0 \rightsquigarrow x = 1 \qquad A = (1;0)$

(3) Segno

$e^x \geq 0$ Sempre

$x - 1 \geq 0 \quad x \geq 1$

$x^2 > 0 \quad x \neq 0$

		0		1	
N_1	+		+		+
N_2	−		−	●	+
D	+	○	+		+
	−		−		+

$\boxed{x \geq 1}$

(4) Limiti: $\lim_{x \to +\infty} e^x \frac{x-1}{x^2} = +\infty$

(H) $\lim_{x \to +\infty} \frac{e^x(x-\cancel{1}) + e^x(\cancel{1})}{2x} = \lim_{x \to +\infty} \frac{e^x \cdot \cancel{x}}{2\cancel{x}} = +\infty$

No Asint. orizz.

$\lim_{x \to +\infty} \frac{e^x(x-1)}{x^2} \cdot \frac{1}{x} = \infty$ Perché e^x più potente $\rightsquigarrow$ No asint. obl.

$\lim_{x \to -\infty} e^x \frac{x-1}{x^2} = 0 \rightsquigarrow$ Asintoto orizzontale

$\lim_{x \to 0^+} e^x \frac{x-1}{x^2} = -\infty \qquad \lim_{x \to 0^-} e^x \frac{x-1}{x^2} = -\infty$

5) Derivata $\quad f(x) = e^x \cdot \frac{x-1}{x^2}$

$$f'(x) = e^x \left(\frac{x-1}{x^2}\right) + e^x \frac{1 \cdot x^2 - (x-1) \cdot 2x}{x^4}$$

$$= e^x \frac{x^3 - \cancel{x^2} + \cancel{x^2} - 2x^2 + 2x}{x^4} = \frac{e^x}{x^4}(x^3 - 2x^2 + 2x)$$

$$= \frac{e^x}{x^3}(x^2 - 2x + 2) \qquad \frac{\Delta}{4} = 1 - 2 = -1$$

$f'(x) \geq 0$

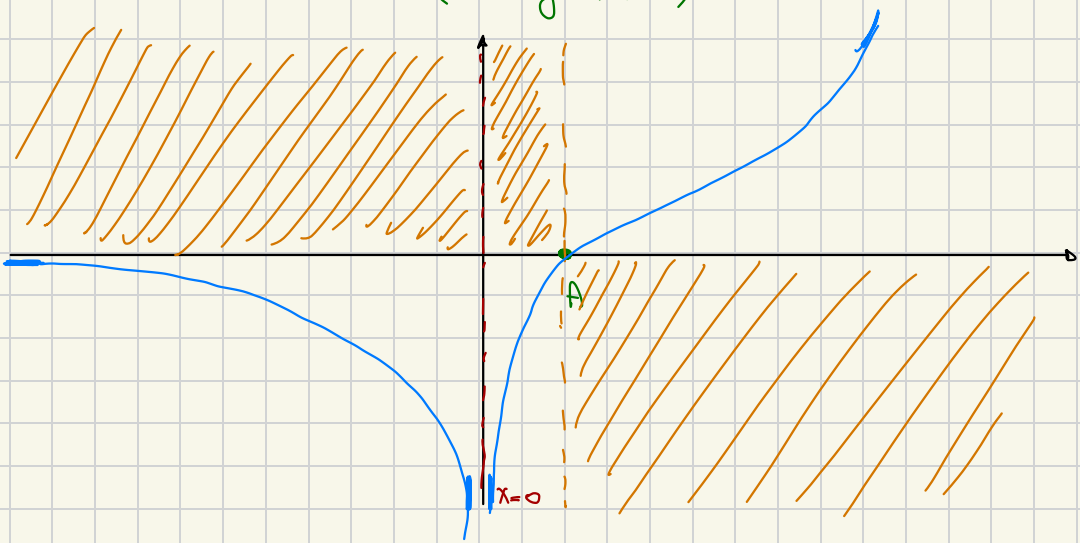
$e^x \geq 0$ sempre

$x^3 > 0 \quad x > 0$

$x^2 - 2x + 2 \geq 0$ sempre

	0	
+		+
−		+
+		+
−		+
↘		↗

"0 è un finto minimo" (Robbby 3/3/26)



(b) $x^3 - 3x^2 + 6x - 6 = 0$ prendi $x \in (1;2)$. Mostra che il polinomio si annulla in $x = \omega$ con ω pto di flesso di $f(x)$

$$f'(x) = e^x \cdot \frac{(x^2 - 2x + 2)}{x^3}$$

$$f''(x) = e^x \frac{x^2 - 2x + 2}{x^3} + e^x \frac{(2x-2)x^3 - 3x^2(x^2 - 2x + 2)}{x^6}$$

$$= \frac{e^x}{x^6}\left[x^5 - 2\cancel{x^4} + 2\cancel{x^3} + 2\cancel{x^4} - 2\cancel{x^3} - 3x^4 + 6x^3 - 6x^2\right]$$

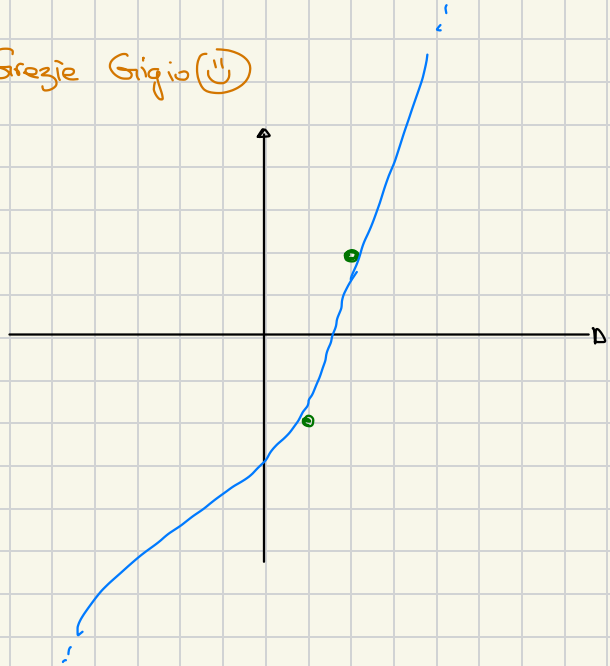
$$= \frac{e^x}{x^{\cancel{6}\,4}} \cancel{x^2} \left(x^3 - 3x^2 + 6x - 6 \right) = \frac{e^x}{x^4} \left(x^3 - 3x^2 + 6x - 6 \right)$$

Provo a usare il teorema degli zeri: $p(x) = x^3 - 3x^2 + 6x - 6$

$p(1) = 1 - 3 + 6 - 6 = -2$

$p(2) = 8 - 12 + 12 - 6 = 2$

Grazie Gigio (☺)



$p'(x) = 3x^2 - 6x + 6$

$p'(x) = 3(x^2 - 2x + 2)$

$\Rightarrow p'(x) \geq 0$ sempre

cioè il polinomio

è sempre crescente